

Quickstart Guide: pmsimstats2025

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Overview

pmsimstats2025 is an N-of-1 clinical trial simulation study comparing multiple trial designs with a focus on statistical power when biomarker- treatment interactions exist and carryover effects are present.

The project implements methodology from **Hendrickson et al. (2020)** with extensions for explicit biomarker moderation and carryover effect modeling.

Installation

```
# Clone the repository
# git clone https://github.com/rgt47/pmsimstats2025.git
# cd pmsimstats2025

# Restore R environment
renv::restore()
```

Quick Start

Run a Simulation

The simulation is organized into two main scripts based on measurement schedule type:

```
# Clustered measurement schedule designs (Hybrid, OL+BDC)
source("analysis/scripts/simulation_clustered.R")

# Evenly-spaced designs (OL, Crossover, Parallel)
source("analysis/scripts/simulation_evenly_spaced.R")
```

Generate Visualizations

```
# Power heatmaps from simulation results
source("analysis/scripts/visualize_hendrickson_style.R")
source("analysis/scripts/visualize_power_heatmaps_4panel.R")
```

Trial Designs

The project supports five trial designs:

Design	Points	Structure
Hybrid	8	Dense cluster at treatment transition
OL+BDC	8	Dense cluster at biomarker discontinuation
Open-Label	4	Evenly spaced (6-week gaps)

Design	Points	Structure
Crossover	4	Evenly spaced (6-week gaps)
Parallel	4	Evenly spaced (6-week gaps)

Measurement Schedule Rationale

Clustered schedules enable more measurement points (8) within a 20-week trial by grouping observations around key transition times. Evenly-spaced designs are limited to 4 points to maintain positive definite covariance matrices.

Core Functions

Parameter Grid Validation

Before running simulations, validate parameter combinations:

```
source("analysis/scripts/pm_functions.R")

# Define parameter grid
param_grid <- expand_grid(
  biomarker_moderation = c(0.2, 0.4),
  carryover_t1half = c(0, 1.0, 2.0),
  n_participants = 70
)

# Validate all combinations
validation <- validate_parameter_grid(param_grid)
report_parameter_validation(validation)
```

Build Sigma Matrix

```
# Construct covariance matrix for a trial design
sigma <- build_sigma_matrix(
  n_timepoints = 8,
  c.br = 0.75,
  c.tr = 0.75,
  c.er = 0.75
)
```

Generate Trial Data

```
# Generate data for one participant
trial_data <- generate_data(
  sigma = sigma,
  n_timepoints = 8,
  treatment_schedule = c(0, 0, 1, 1, 1, 1, 0, 0)
)
```

Carryover Modeling

The simulation compares two analysis scenarios:

With Explicit Carryover

```
library(lme4)

fit_with <- lmer(
  response ~ treatment * biomarker + week + carryover_effect +
    (1 | participant_id),
  data = trial_data
)
```

Without Carryover (Hendrickson Original)

```
fit_without <- lmer(
  response ~ treatment * biomarker + week + (1 | participant_id),
  data = trial_data
)
```

Carryover Computation

Carryover effect is modeled as exponential decay:

```
carryover_effect <- scale_factor * (0.5)^(weeks_off / t1half)
```

Where:

- `scale_factor` = 1.0 (BR-only carryover)
- `t1half` is the half-life in weeks

Output Files

Simulation Results

Results are saved as RData files:

- `analysis/output/simulation_clustered_results.RData`
- `analysis/output/simulation_evenly_spaced_results.RData`

Each file contains:

- `simulation_results` - One row per iteration with p-values and effect sizes
- `simulation_summary` - Aggregated power by condition
- `param_grid` - Parameter specifications

Visualizations

Power heatmaps are saved as PDF files:

- `analysis/output/power_heatmap_clustered.pdf`
- `analysis/output/power_heatmap_evenly_spaced.pdf`
- `analysis/output/power_heatmap_4panel.pdf`

Docker Workflow

For reproducibility, run simulations in Docker:

```
# Start container with all dependencies
make r
```

```
# Inside container, run simulations
Rscript analysis/scripts/simulation_clustered.R
```

Modifying Parameters

Change Biomarker Correlation Grid

Edit the parameter grid in simulation files:

```
param_grid <- expand_grid(
  biomarker_moderation = c(0.2, 0.4, 0.6),
  carryover_tihalf = c(0, 1.0, 2.0),
  n_participants = 70
)
```

Adjust Iteration Count

Change `n_iterations` in simulation files:

```
n_iterations <- 100 # More stable estimates (slower)
```

Troubleshooting

Non-Positive Definite Matrix

If you see “Non-positive definite matrix rejected”, this is normal behavior for invalid parameter combinations. Check:

1. Biomarker correlation values (>0.4 may cause issues)
2. Measurement schedule spacing
3. Condition number output

Convergence Warnings

Occasional mixed model convergence warnings are expected. Monitor the **error** column in results if >20% of iterations fail.

References

Hendrickson, E., et al. (2020). N-of-1 trials with multiple randomization structures for individualized treatment. *Statistics in Medicine*, 39(25), 3581-3599.

Next Steps

- analysis/scripts/README.md - Detailed workflow guide
- docs/simulation_white_paper.md - Methodology documentation
- docs/sigma_matrix_derivation.tex - Mathematical foundations